

Transfusion medicine

Products, thresholds and special processing · reactions · blood bank workup · refractoriness, massive hemorrhage, apheresis · **pages 1–2 on-call consult, page 3 board reference**

Compiled from the Fellowship Transfusion Medicine review, The Fellow on Call ep. 060, and AABB / ASFA / ISBT guidance · primary sources verified against PubMed, October 2026.

THE CONSULT — ESTABLISH FIRST

- **Transfuse for a physiologic reason, not a number.** Trend, symptoms, bleeding and comorbidity outweigh the absolute value; a stable Hb of 6.8 from iron deficiency may need IV iron, not blood
- **What is the deficit?** Hb, platelets, PT/INR + aPTT, **fibrinogen**, a specific factor
- **Bleeding or stable?** Thresholds below are for **non-bleeding** patients; active hemorrhage is treated to clinical endpoints
- **Blood bank history:** prior **alloantibodies**, prior reactions, **anti-CD38 therapy**, transplant (ABO switch), pregnancy
- **Special requirements:** irradiated, antigen-negative, HLA-matched, washed (table at right)
- **Goals / refusal:** ask product by product (many who decline RBCs accept albumin, factor concentrates, cell salvage); plan IV iron, ESA, TXA, minimal phlebotomy
- **Single-unit policy:** 1 RBC unit (or 1 apheresis platelet) → reassess before the next. Most over-transfusion is a reflexive 2-unit order
- **Stanford:** Transfusion Medicine pager **12027** · Apheresis pager **19164** (650-725-4656) · Apheresis fellow **27283** · Special Coag lab 650-723-4813

PRODUCTS — WHAT YOU GET

Product	Contents · adult dose → effect
RBCs	~300 mL, Hct ~55–65%. 1 unit → Hb ↑ ~1 g/dL (Hct ~3%) in a non-bleeding adult
Platelets	1 apheresis unit ≈ a pool of 4–6 whole-blood-derived; ≥3×10¹¹ . 1 unit → ↑ ~30k (less if refractory, febrile, splenomegalic). Half-dose = full dose for bleeding (PLADO)
Plasma (FFP / PF24)	All coagulation factors. 10–15 mL/kg (~4 units in 70 kg) to correct a real coagulopathy
Cryo	Fibrinogen , FVIII, vWF, FXIII, fibronectin. 10 units (2 pre-pooled bags of 5) → fibrinogen ↑ ~70–100 mg/dL
4F-PCC	II, VII, IX, X + proteins C/S. Warfarin + major bleed: 25 / 35 / 50 units/kg for INR 2–<4 / 4–6 / >6 + vitamin K 10 mg IV

RBC THRESHOLDS — AABB 2023

Setting	Transfuse at Hb
Hospitalized, hemodynamically stable (incl. ICU)	<7 g/dL <i>strong rec.</i>
Heme/onc inpatients (chemo, HSCT)	<7 g/dL <i>conditional, low certainty</i>
Cardiac surgery	<7.5 g/dL
Orthopedic surgery or pre-existing CVD	<8 g/dL (FOCUS)
Septic shock	<7 g/dL (TRISS)
Acute MI + anemia	Liberal <10 favored/considered (MINT) — MI or death 16.9% vs 14.5%, RR 1.15, P=0.07 (not significant) ; not covered by AABB 2023
Active bleeding / shock	Clinical endpoints, not a number

Fresh blood is not better: <8-day vs standard-issue RBCs did not change mortality (ABLE; RECESS) — don't request "fresh" units.

PLATELET THRESHOLDS — AABB/ICTMG 2025

Setting	Transfuse below
Chemo or allo-HSCT, not bleeding	10k <i>strong</i>
Auto-HSCT or aplastic anemia, not bleeding	No routine prophylaxis <i>conditional</i>
Consumptive (adult), no major bleeding	10k
Fever / sepsis	~ 20k <i>convention (Rebulla protocol)</i>
CVC, compressible site	10k <i>was 20k in 2015</i>
Lumbar puncture	20k <i>strong; was 50k in 2015</i>
IR procedure: low / high risk	20k / 50k
Major non-neuraxial surgery	50k
Neurosurgery, ocular, neuraxial	~ 100k <i>convention</i>
Active bleeding / CNS or massive bleed	> 50k / >100k <i>convention</i>
ICH with plt >100k, incl. on antiplatelets	Do not transfuse

WHEN NOT TO GIVE PLATELETS

- **TTP and HIT:** avoid unless life-threatening bleeding — fuels thrombosis (observational data)
- **ITP:** consumed within hours; only for serious bleeding, **with IVIG + steroids**
- **Post-transfusion purpura:** ineffective → **IVIG**
- Non-thrombocytopenic cardiac surgery without major bleeding

PLASMA · CRYO · REVERSAL

- **Plasma for:** bleeding + coagulopathy (liver, DIC, dilution), MTP, **TTP plasma-exchange replacement**, factor deficiency without a concentrate (FV)
- **Plasma misuse: don't correct INR <1.7 before a procedure** — it barely moves and adds TACO/TRALI risk; never as a volume expander; **not for warfarin reversal when PCC is available**
- **Cryo for fibrinogen <150 mg/dL with bleeding** (DIC, MTP, APL) — or fibrinogen concentrate
- **Dabigatran:** idarucizumab 5 g IV. **Xa inhibitors:** 4F-PCC (off-label) — **andexanet left the US market Dec 2025**

SPECIAL PROCESSING — WHO NEEDS WHAT

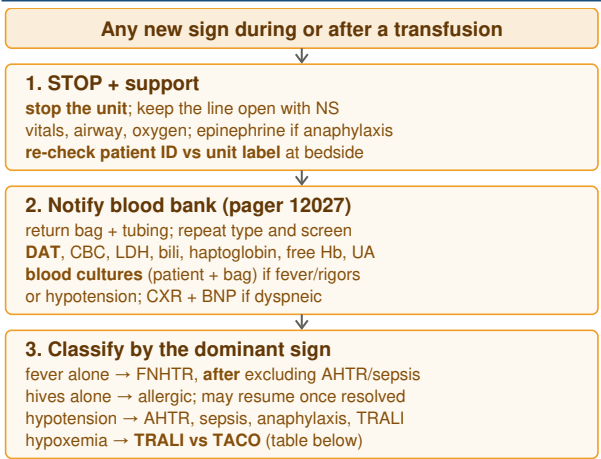
Modification	Does · who needs it
Irradiated <i>25 Gy</i>	Inactivates donor T cells → prevents TA-GVHD. Allo-HSCT (from conditioning while immunosuppressed) · auto-HSCT (7 d before harvest → 3 mo post) · Hodgkin lymphoma, any stage — indefinitely · purine analogs: fludarabine, cladribine, pentostatin, bendamustine — indefinitely · alemtuzumab · congenital T-cell immunodeficiency · IUT/neonatal exchange · HLA-matched or relative-donated units , granulocytes · CAR-T per institutional policy. RBC shelf life drops to 28 d, K ⁺ rises
Pathogen-reduced plt	Amotosalen/UVA — accepted alternative to irradiation ; also mitigates bacterial risk
Leukoreduced	<5×10 ⁶ WBC/unit. ↓ FNHTR, ↓ HLA alloimmunization/refractoriness (TRAP), ↓ CMV. Most US units are pre-storage LR
CMV-safe	LR ≈ CMV-seronegative for HSCT, pregnancy, fetal/neonatal — either is acceptable
Washed	Removes plasma: recurrent severe allergic reactions , anaphylaxis with anti-IgA (or IgA-deficient donor), hyperkalemia risk
Antigen-negative	Known alloantibody; sickle cell: Rh (C, E) + K matched ; anti-CD38 therapy (K-negative)
HLA-matched / crossmatched plt	Immune refractoriness (below). Irradiate

PLATELET REFRACTORINESS

- **CCI** = (post – pre count/μL) × BSA (m²) ÷ platelets given (×10¹¹). **1-h CCI <5,000–7,500 on 2 consecutive ABO-compatible units** = refractory
- **Good 1-h, poor 24-h** → **non-immune** (most common): fever, sepsis, DIC, splenomegaly, bleeding, drugs (amphotericin), VOD/TA-TMA
- **Poor 1-h** → **immune:** mostly **HLA class I** (HPA less often) → send **HLA class I antibody screen** → **HLA-matched, antigen-avoidant, or crossmatch-compatible** platelets, ABO-identical, irradiated
- Also think drug-induced immune thrombocytopenia, HIT, PTP. Prevention = leukoreduction (TRAP: 13% → 3–5%)
- Bleeding while refractory: keep transfusing + **antifibrinolytic** (TXA)

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ACUTE REACTION — FIRST 15 MINUTES



MASSIVE HEMORRHAGE

- **Activate per institutional MTP**; send T&S, CBC, PT/aPTT, fibrinogen, ionized Ca, ABG, lactate
- **1:1:1 plasma : platelets : RBC** (PROPPR vs 1:1:2: 24-h and 30-d mortality **not different**, but **hemostasis 86% vs 78%** and **fewer exsanguination deaths at 24 h, 9.2% vs 14.6%**)
- **TXA 1 g over 10 min + 1 g over 8 h, within 3 h** of trauma (CRASH-2: mortality RR 0.91). **After 3 h, bleeding deaths rose (RR 1.44) — don't start late**
- **Calcium** (citrate): follow iCa, replace; prevent **hypothermia** (warmer), acidosis; watch K⁺
- **Fibrinogen <150 → cryo**; viscoelastic testing (TEG/ROTEM) where available; reverse anticoagulants

ORDERING AND THE BEDSIDE

- **Two-identifier bedside check** of patient vs unit — the single most important ABO safeguard
- **Only normal saline in the line** — LR's calcium clots citrated blood; hypotonic fluids hemolyze; no co-infused drugs
- **Finish each unit within 4 h** of issue (bacterial growth). Typical RBC unit over 1–2 h; ~1 mL/kg/h if TACO-prone (HF, CKD, elderly, positive balance)
- **No routine premedication** — acetaminophen/diphenhydramine did not reduce reactions in randomized trials; reserve for recurrent reactors
- Consent covers reactions, TACO/TRALI and (very low) infection risk; document alternatives offered

REACTIONS

Reaction	Timing · hallmark	Management · prevention
AHTR	Minutes. ABO mismatch (clerical) → intravascular hemolysis: fever, flank pain, hemoglobinuria, shock, DIC, AKI	Stop; fluids for urine output, pressors, treat DIC. Two-identifier bedside check
FNHTR	During/<4 h; fever ± rigors; cytokines/anti-WBC	Antipyretic; diagnosis of exclusion. LR prevents
Allergic	Urticaria, pruritus only	Pause, antihistamine, resume. No routine premed
Anaphylaxis	Seconds–minutes; hypotension, bronchospasm. Anti-IgA in IgA deficiency (or haptoglobin deficiency)	Epinephrine. Check IgA; then washed or IgA-deficient products
Septic	High fever, rigors, shock; platelets (room temp) highest risk	Cultures (patient + bag), broad antibiotics; tell blood bank to quarantine co-components
Hypotensive	Isolated ↓BP within minutes; ACE inhibitors (bradykinin)	Resolves on stopping; hold ACE-i
DHTR	3–14 d. Anamnestic alloantibody (Kidd classic) → Hb drop, ↑bili, DAT+ (IgG) , new antibody	Usually supportive; antigen-negative units going forward
Hyperhemolysis	SCD: Hb falls below pre-transfusion ; own + donor cells lysed, often DAT–	Stop transfusing ; IVIG + steroids ± eculizumab
TA-GVHD	1–6 wk. Fever, rash, diarrhea, LFTs, marrow aplasia ; >90% fatal	Prevention only: irradiate or pathogen-reduce
PTP	~5–10 d. Severe thrombocytopenia; anti-HPA-1a , classically multiparous women	IVIG ; platelets ineffective; HPA-1a– if needed
Iron overload	Chronic transfusion (>~20 units) — heart, liver, endocrine	Ferritin, cardiac/liver T2* MRI; chelate

APHERESIS — ASFA 2023 (PAGER 19164)

Indication	Procedure · category
TTP	Daily TPE — Cat I ; start on suspicion (PLASMIC), don't wait for ADAMTS13. + steroids + caplacizumab (HERCULES: recurrence 12% vs 38%) ± rituximab
Hyperviscosity (symptomatic)	TPE — Cat I (Waldenström, myeloma)
Leukostasis	Leukocytapheresis — Cat III, not routine ; prophylactic also III. Cytoreduce (hydroxyurea, induction). Avoid in APL
Sickle cell	RBC exchange: acute stroke Cat I , stroke prophylaxis I, acute chest II . Target HbS <30% ; avoid Hb >10–11
Other Cat I TPE	GBS, myasthenia gravis, CAPS, anti-GBM, ABO-incompatible / HLA desensitization (living-donor kidney)

BLOOD BANK WORKUP

- **Type and screen** = ABO/RhD + **antibody screen** (patient plasma vs reagent O cells, indirect antiglobulin test). Order it early in anyone likely to bleed
- **Crossmatch: electronic** (screen negative + 2 concordant ABO types) → minutes. **Full AHG crossmatch** if an alloantibody is present
- **Positive screen → antibody ID panel → antigen-negative units** — hours to days for multiple/rare antibodies. **Call early** in SCD, thalassemia, multiparous, chronically transfused
- **DAT** (patient RBCs): **IgG ± C3** → warm AIHA, DHTR, drug; **C3 only** → cold agglutinin, PCH
- **Warm AIHA**: panreactive autoantibody; blood bank adsorbs to exclude hidden alloantibodies, then issues “**least incompatible**” units — **never withhold blood in life-threatening anemia**
- **Daratumumab / isatuximab** (CD38 on reagent RBCs) → **panreactive screen**. Before first dose: **type, screen + phenotype/genotype**, tell the blood bank. Lab uses **DTT-treated cells** → DTT destroys Kell, so give **K-negative units**
- **Emergency release**: uncrossmatched **O-neg** for females of childbearing potential and unknowns; **O-pos** for men and older women; **AB plasma**. Switch to type-specific ASAP
- **RhD-neg female of childbearing potential given RhD-pos platelets** → **RhIG** (residual RBCs can immunize)
- **ABO-mismatched HSCT**: blood bank picks the product group by transplant phase — don't override

TRALI VS TACO

	TRALI	TACO
Definition	Within 6 h ; ARDS-like, no LAH. Type I (no ARDS risk factor) / II <i>Vlaar 2019</i>	Within 12 h ; respiratory distress or pulmonary edema + ≥3 criteria <i>ISBT 2018</i>
Mechanism	Donor anti-HLA / anti-HNA → capillary leak	Hydrostatic volume overload
BP / JVP	Hypotension ; JVP normal	Hypertension ; JVP ↑
Fever	Often	Usually absent
BNP	Normal	↑ (NT-proBNP supports)
Diuretic	No response — don't	Improves
Prevent	Male/never-pregnant plasma donors	Slow rate, single units, pre-emptive diuretic in at-risk

Report both to the blood bank. TRALI does not require antibody detection to diagnose.

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BLOOD GROUPS AND ANTIBODIES

System	Antibody · clinical significance	Board pearl
ABO	Naturally occurring IgM anti-A/ anti-B (group O also IgG) → intravascular AHTR	O = universal RBC donor; AB = universal plasma donor ; plasma compatibility runs in reverse
Rh (D, C, c, E, e)	IgG; D most immunogenic . DHTR, severe HDFN	RhIG prevents anti-D only; passive anti-D makes the screen positive
Kell (K)	IgG; DHTR, HDFN via erythroid suppression (anemia > bilirubin)	~91% of donors K-negative; McLeod (absent Kx) → acanthocytes
Duffy (Fy ^a / Fy ^b)	IgG; DHTR, HDFN	Fy(a–b–) common with African ancestry → P. vivax resistance + Duffy-null associated neutrophil count (benign low ANC)
Kidd (Jk ^a /Jk ^b)	IgG, complement-fixing; titers evanesce below detection	Classic cause of DHTR — screen negative, then anamnestic boost
MNS	Anti-M/N usually cold IgM (insignificant); anti-S, s, U are significant	U-negative units are rare (African ancestry)
Lewis	IgM, insignificant; antigens adsorbed from plasma	Common in pregnancy; no HDFN
P	Anti-P1 insignificant. Donath–Landsteiner : biphasic IgG anti-P	PCH — children after viral illness; DAT C3+
I / i	Cold IgM autoantibodies	Anti-I: Mycoplasma , primary CAD · anti-i: EBV , lymphoma

Clinically significant = IgG reacting at 37°C/AHG (Rh, Kell, Duffy, Kidd, S/s). Usually insignificant = IgM cold (Lewis, M, N, P1, I). Mnemonic: “Kidd kills, Duffy dies, Lewis lives.”

DAT PATTERNS

Result	Think
IgG ± C3	Warm AIHA (CLL, SLE, drugs), DHTR (mixed-field: only donor cells coated), HDFN, drug (penicillin, cephalosporins)
C3 only	Cold agglutinin disease (IgM, Mycoplasma, LPL/ Waldenström), PCH , immune-complex drug
Negative + hemolysis	Spherocytosis, G6PD, MAHA, PNH ; DAT-negative AIHA (low-affinity IgG, IgA)
Positive, no hemolysis	IVIg, ATG, recent RhIG; ~1 in several thousand healthy donors

ABO COMPATIBILITY

Recipient	RBCs	Plasma / cryo
O	O only	O, A, B, AB
A	A, O	A, AB
B	B, O	B, AB
AB	A, B, AB, O	AB only

RBCs: donor cells must lack the antigen the recipient's antibody sees. Plasma: the reverse. **Platelets** are given ABO-compatible when possible (out-of-group plasma can hemolyze); **ABO-identical improves increments** in refractoriness).

HDFN AND RH IMMUNE GLOBULIN

- **RhD-neg mother**: RhIG **300 µg at 28 wk + within 72 h of delivery** of an RhD-pos baby, and after bleeding/procedures
- **One 300-µg vial covers 30 mL fetal whole blood (15 mL fetal RBCs)**
- **Rosette screen** → **Kleihauer–Betke** (or flow) to size the bleed: fetal blood (mL) ≈ KB% × 50; vials = mL ÷ 30, round up + 1
- Once alloimmunized, RhIG is useless → titers, then **MCA Doppler**. **Anti-K: titers unreliable** — follow by Doppler
- IV anti-D for ITP needs RhD-pos, spleen intact; risk of intravascular hemolysis

STORAGE AND INFECTION RISK

Product	Storage · shelf life	Pearl
RBCs	1–6°C; 42 d (additive solution); 28 d once irradiated	Storage lesion: K ⁺ leak, ↓2,3-DPG — age did not matter clinically (ABLE, RECESS)
Platelets	20–24°C with agitation; 5 d (7 d with FDA-approved bacterial testing)	Highest bacterial risk — FDA 2019 guidance: LVDS, secondary testing or pathogen reduction
Plasma	Frozen ≤–18°C, 1 yr ; thawed 24 h (thawed plasma 5 d at 1–6°C)	Labile FV/FVIII fall after thaw
Cryo	Frozen, 1 yr ; use within 4–6 h of thaw/pool	Room temperature after thaw

Viral residual risk in high-income systems with NAT: **HIV and HCV** ~<1 per 1–2 million units; HBV somewhat higher. **Non-infectious risks** (TACO, TRALI, hemolysis) now dominate transfusion deaths.

KEY TRIALS

Trial	Result → practice
TRICC 1999	ICU Hb 7 vs 10: 30-d mortality 18.7% vs 23.3% (P=0.11) → restrictive is safe
FOCUS 2011	Hip fracture + CV risk, Hb 10 vs symptoms/<8: death or inability to walk 35.2% vs 34.7% → restrictive OK in CVD
TRISS 2014	Septic shock, Hb 7 vs 9: 90-d mortality 43.0% vs 45.0% (RR 0.94) → 7 in sepsis
MINT 2023	Acute MI, liberal <10 vs restrictive 7–8 (n=3,504): MI/death 14.5% vs 16.9%, RR 1.15 (0.99–1.34), P=0.07 ; death 8.3% vs 9.9% → liberal favored, not proven
ABLE / RECESS 2015	Fresh (<8 d / ≤10 d) vs standard RBCs: no mortality or organ-dysfunction difference → don't request fresh units
Rebulla 1997	AML induction, plt threshold 10k vs 20k: major bleeding 21.5% vs 20%, 21.5% fewer platelet transfusions → the 10k threshold
PLADO 2010	Platelet dose (1.1 / 2.2 / 4.4 ×10 ¹¹ /m²) at ≤10k: grade ≥2 bleeding 71 / 69 / 70% → dose doesn't matter; a dose trial, not a threshold trial
TOPPS 2013	No-prophylaxis vs prophylaxis <10k: bleeding 50% vs 43%, non-inferiority not shown → keep prophylaxis after chemo/allo
TRAP 1997	AML: leukoreduced or UVB platelets cut alloimmune refractoriness 13% → 3–5%
PROPPR 2015	Trauma 1:1:1 vs 1:1:2: 30-d mortality 22.4% vs 26.1% (P=0.26); hemostasis 86% vs 78%; exsanguination death 9.2% vs 14.6%
CRASH-2 2010/11	Trauma TXA: all-cause death 14.5% vs 16.0% (RR 0.91); benefit only if given ≤3 h
HERCULES 2019	iTTP caplacizumab + TPE: composite TTP death/recurrence/TE 12% vs 49%; recurrence 12% vs 38%

SICKLE CELL AND CHRONIC TRANSFUSION

- **Prophylactic Rh (C, E) + K antigen matching** for all SCD transfusions; extended matching or RH genotyping if alloimmunized
- **Simple transfusion** for symptomatic anemia; **exchange** to lower HbS (stroke, severe ACS, high-risk surgery). **Never push Hb >10–11** — viscosity
- **DHTR in SCD mimics a pain crisis** — check HbA% and a new antibody; hyperhemolysis = stop transfusing
- **Iron overload**: chelate with deferasirox, deferoxamine or deferiprone (agranulocytosis — weekly ANC)